

Davis Balgley

SENIOR SOFTWARE ENGINEER | DISTRIBUTED SYSTEMS | ML-DRIVEN APPLICATIONS

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SUMMARY

Senior Software Engineer with experience building real-time data processing systems, cloud-native applications, and ML-driven analytics platforms. Background includes distributed systems architecture, Kubernetes-based deployments, and applied machine learning across defense, biotech, and SaaS domains. Proven ability to lead complex projects, modernize legacy systems, and deliver production-ready software in high-stakes environments.

AREAS OF EXPERTISE

Software Languages: Python, JAVA, R Statistical, C++, SQL, R, Bash

Software Dev. Methodologies: Agile (Scrum, Kanban), Git

Domains: Machine Learning, Orbital Dynamics, Signal Processing, Bioinformatics, Natural Language Processing

AWARDS

2017-2018 MIT LINCOLN LABORATORY TEAM AWARD FOR
SIGNIFICANT CONTRIBUTIONS TO SPACE SITUATIONAL AWARENESS

CREDENTIALS

DEPARTMENT OF DEFENSE TOP SECRET ACTIVE 2018
AWS CERTIFIED PRACTITIONER 2025

PROFESSIONAL EXPERIENCE



OUTSIDE ANALYTICS, INC./SMX, INC.
Broomfield, Colorado
Shared Vision, Outcome Assured.

JUL 2023 TO CURRENT (2YR. 9 MO.)

SENIOR SOFTWARE ENGINEER (Python, Kubernetes, AWS, PostgreSQL) – Contributed to a software incubation program modernizing legacy sensor processing systems through containerization and cloud-native deployment. Developed and maintained a real-time processing framework deployed on Rancher Kubernetes and AWS EKS.

- Led redesign of a legacy sensor processing framework, including data flow architecture, unit-of-work modeling, containerization, and Kubernetes deployment. Enabled deployment to air-gapped environments and supported adoption across multiple external organizations. Eliminated over 25,000 SLOC during the redesign, improving developer experience and streamlining a major repository.
- Improved a CNN-based real-time sensor analysis application, increasing true positive detection rates by 30% through model refinement and pipeline optimization. Built supporting tools for output analysis and visualization; system deployed across multiple production pipelines.
- Built developer tooling and simulation infrastructure supporting mission planning and pre-visualization workflows, including a mission editor and scenario generation tools used in advanced R&D environments.
- Collaborated with multiple teams to integrate sensor processing capabilities (format adapters, data pipelines, algorithms) into broader organizational systems, improving reuse and developer adoption.
- Mentored junior engineers in sensor processing systems and domain concepts, improving onboarding and development velocity. Contributed to engineering standards and best practices for framework development.



ARCHERDX, INC./INVITAE CORP./IDT, INC.
Boulder, Colorado
From Genetics, Health.

SEP 2020 TO JUL 2023 (2YR. 10 MO.)

BIOINFORMATICS SOFTWARE ENGINEER III – TECHNICAL LEAD (Python, Docker, AWS, PostgreSQL) – Led development of bioinformatics pipelines for cancer diagnostics following successive acquisitions (Invitae and Integrated DNA Technologies). Maintained delivery continuity and adapting processes, architecture, and team structure during organizational transition.

- Led integration of engineering practices and systems post-acquisition, including process standardization, documentation, and technical transition planning.
- Owned architecture and performance improvements for variant calling pipelines, including algorithmic enhancements to increase sensitivity.
- Modernized pipeline infrastructure using containerized services, introducing validation frameworks to ensure data parity, improved accuracy, and reduced runtime and resource utilization.

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- Selected to lead a high-risk program, managing a team of 10 engineers to deliver a critical software release on schedule; product supported >\$10M in revenue in FY22.
- Served as primary technical authority, advising stakeholders and leadership on architecture, tradeoffs, and implementation strategy.
- Presented timelines, risks, and resourcing plans to senior and executive leadership.
- Led release planning and execution, coordinating cross-team readiness.
- Introduced engineering initiatives to improve engagement, code quality, and retention.



COLLECTIVE INTELLIGENCE, INC.
Lakewood, Colorado
Crowdsource Better Projects.

JUN 2019 TO JUL 2020 (1 YR. 1 MO.)

SENIOR SOFTWARE ENGINEER (Python Flask, VueJS, PostgreSQL) - Part of the founding engineering team responsible for full-stack development and product feature design across backend (Flask) and frontend (VueJS). Focused on business logic, engagement systems, and early-stage ML/NLP exploration for user feedback analysis and product insights.

- Designed and implemented text scoring algorithms to structure and prioritize user feedback for content creators. Presented as flagship feature for product.
- Built tool for collaborative content creation workflows enabling real-time, feedback-driven editing for articles and blog posts.
- Applied NLP techniques using spaCy to extract features and generate structured insights from unstructured text data.
- Influenced product and system design by evaluating tradeoffs between implementation complexity, performance, and user experience in a resource-constrained startup environment.
- Partnered with CEO and executive team to prioritize and deliver high-impact product features.



MIT LINCOLN LABORATORY
Colorado Springs, Colorado & Lexington, Massachusetts
Technology in Support of National Security.

FEB 2017 TO MAY 2019 (2 YRS. 4 MO.)

SOFTWARE ENGINEER (Java, PostgreSQL, JBoss Fuse ESB, Camel, Apache/AMQ, MATLAB, Deep Learning) - Developed image processing and data analysis systems for space-based sensing platforms in a government R&D environment. Contributed to a shared image processing framework supporting multiple sensing platforms. Responsibilities included software development, project scoping, stakeholder communication, and field support across both new development and legacy system sustainment.

- Applied cNN-based approaches to automate feature recognition in imagery, reducing reliance on manual analysis.
- Designed and implemented astrodynamics algorithms for object tracking and classification.
- Designed and implemented a data processing pipeline with integrated metrics and status reporting via RESTful interfaces.
- Developed processing algorithms to improve data efficiency in imaging workflows.
- Delivered rapid-turnaround software updates to support post-launch satellite system tuning.
- Designed and implemented a new image output format aligning legacy systems with updated transmission and reporting standards.



APPLIED RESEARCH ASSOCIATES, INC.
Littleton, Colorado & Arlington, Virginia
Expanding the realm of possibility.

AUG 2014 TO FEB 2017 (2 YRS. 7 MO.)

SOFTWARE ENGINEER (Python, Java, C++, R Statistical, Machine Learning) - Developed statistical and machine learning models for analysis of medical imaging data in a neuroscience research program focused on mild traumatic brain injury (mTBI). Served as primary point of contact for statistical analysis and software design on a cross-functional team of engineers and biomedical researchers.

- Collaborated with leadership to develop repeatable methods for distinguishing between injured and uninjured subjects based on imaging data.
- Developed logistical regression models for analysis of medical imaging data.
- Designed and implemented machine learning models (e.g., SVM, k-means clustering, neural networks) to support classification and pattern detection in imaging data.

EDUCATION



COLORADO SCHOOL OF MINES – Golden, Colorado 2014

BACHELOR OF SCIENCE, COMPUTER SCIENCE WITH A FOCUS IN APPLIED MATHEMATICS AND BIOMEDICAL ENGINEERING